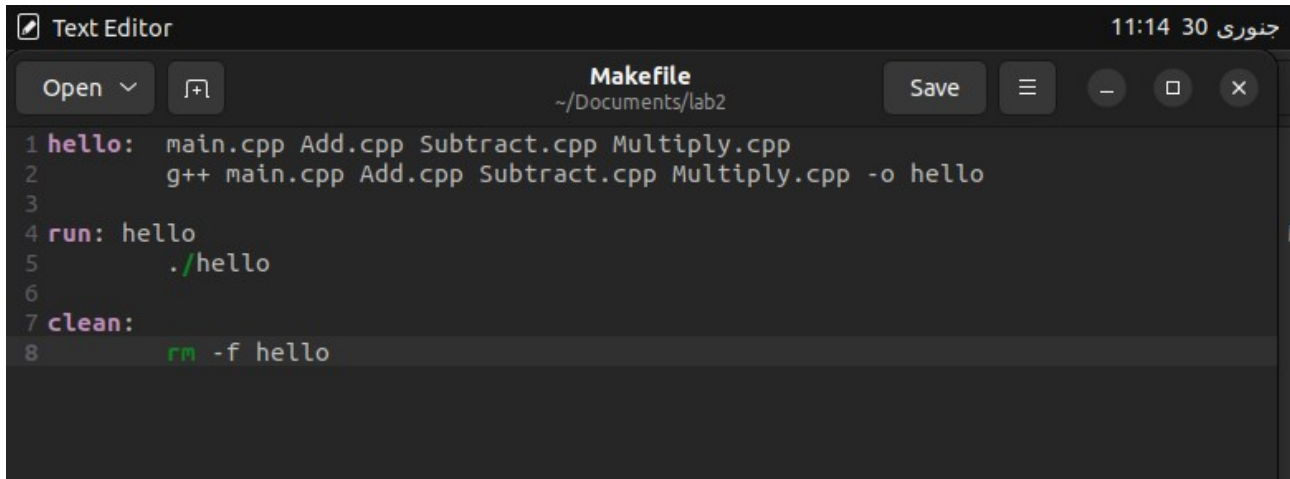


LAB 2 tasks

Task 1:

Make file:



The screenshot shows a Text Editor window with a dark theme. The title bar indicates the file is 'Makefile' located at '~/Documents/lab2'. The editor contains the following content:

```
1 hello: main.cpp Add.cpp Subtract.cpp Multiply.cpp
2       g++ main.cpp Add.cpp Subtract.cpp Multiply.cpp -o hello
3
4 run: hello
5       ./hello
6
7 clean:
8       rm -f hello
```

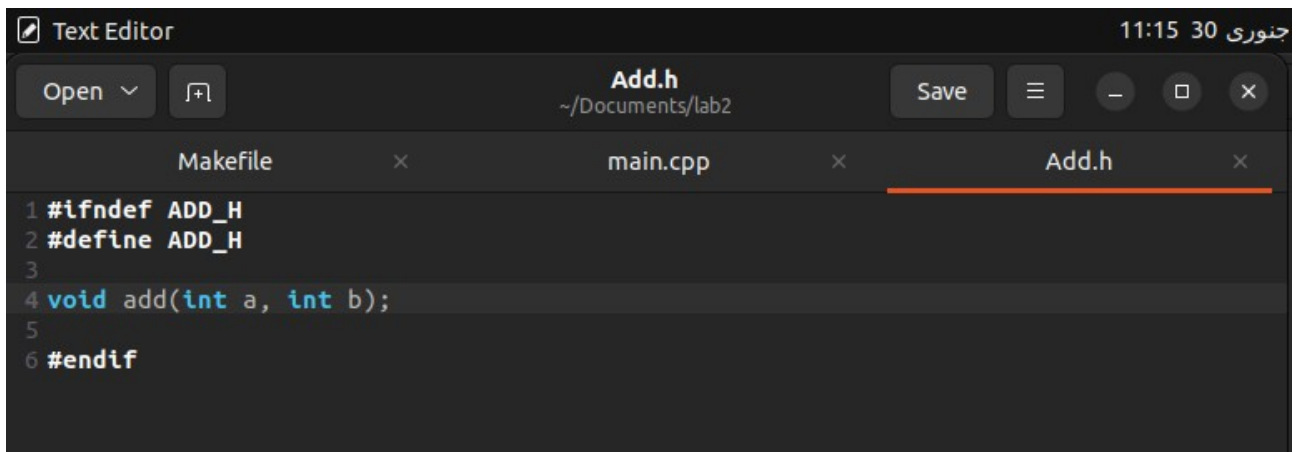
Main.cpp:



The screenshot shows a Text Editor window with a dark theme. The title bar indicates the file is 'main.cpp' located at '~/Documents/lab2'. The editor contains the following content:

```
1 #include <iostream>
2 #include "Add.h"
3 #include "Subtract.h"
4 #include "Multiply.h"
5
6 using namespace std;
7
8 int main() {
9     int a, b;
10
11     cout << "Enter two numbers: ";
12     cin >> a >> b;
13
14     add(a, b);
15     subtract(a, b);
16     multiply(a, b);
17
18     return 0;
19 }
```

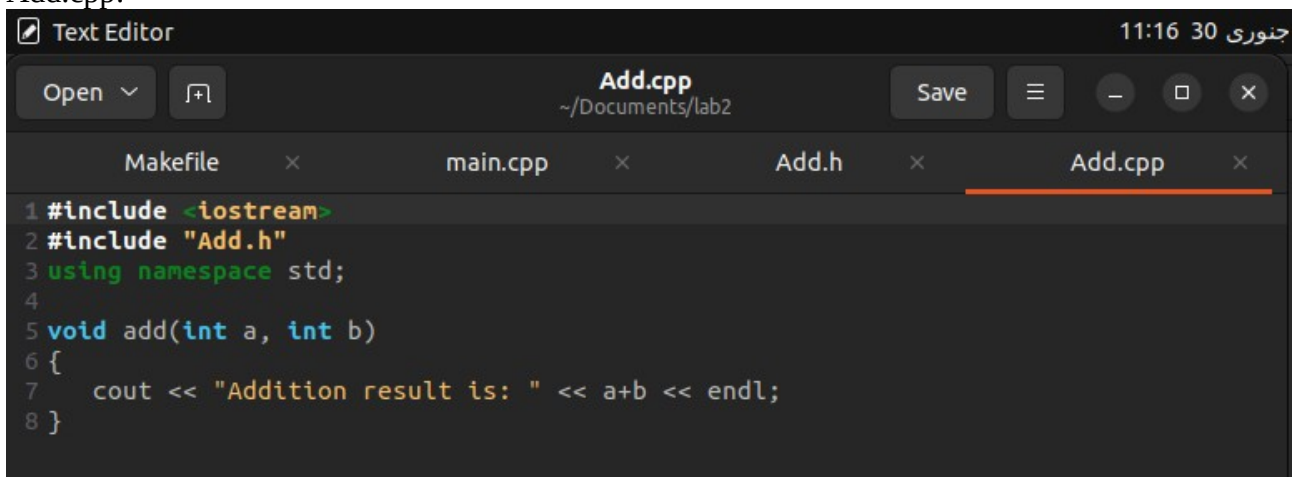
Add.h:



The screenshot shows a Text Editor window with the title bar "Text Editor" and a timestamp "11:15 30 جنوری". The window has a menu bar with "Open", a file icon, and "Save". The title bar also shows the file name "Add.h" and the path "~/Documents/lab2". The editor has three tabs: "Makefile", "main.cpp", and "Add.h". The "Add.h" tab is active and shows the following code:

```
1 #ifndef ADD_H
2 #define ADD_H
3
4 void add(int a, int b);
5
6 #endif
```

Add.cpp:



The screenshot shows a Text Editor window with the title bar "Text Editor" and a timestamp "11:16 30 جنوری". The window has a menu bar with "Open", a file icon, and "Save". The title bar also shows the file name "Add.cpp" and the path "~/Documents/lab2". The editor has four tabs: "Makefile", "main.cpp", "Add.h", and "Add.cpp". The "Add.cpp" tab is active and shows the following code:

```
1 #include <iostream>
2 #include "Add.h"
3 using namespace std;
4
5 void add(int a, int b)
6 {
7     cout << "Addition result is: " << a+b << endl;
8 }
```

Subtract.h:



The screenshot shows a Text Editor window with the title bar "Text Editor" and a timestamp "11:17 30 جنوری". The window has a menu bar with "Open", a file icon, and "Save". The title bar also shows the file name "Subtract.h" and the path "~/Documents/lab2". The editor has three tabs: "Makefile", "main.cpp", and "Subtract.h". The "Subtract.h" tab is active and shows the following code:

```
1 #ifndef SUB_H
2 #define SUB_H
3
4 void subtract(int a, int b);
5
6 #endif
```

Make run;

```
student@student-OptiPlex-7000:~/Documents$ mkdir lab2
student@student-OptiPlex-7000:~/Documents$ cd lab2
student@student-OptiPlex-7000:~/Documents/lab2$ gedit Makefile
student@student-OptiPlex-7000:~/Documents/lab2$ touch Add.cpp
student@student-OptiPlex-7000:~/Documents/lab2$ gedit Add.cpp
student@student-OptiPlex-7000:~/Documents/lab2$ touch Subtract.cpp
student@student-OptiPlex-7000:~/Documents/lab2$ gedit Subtract.cpp
student@student-OptiPlex-7000:~/Documents/lab2$ touch Multiply.cpp
student@student-OptiPlex-7000:~/Documents/lab2$ gedit Multiply.cpp
student@student-OptiPlex-7000:~/Documents/lab2$ touch Add.h
student@student-OptiPlex-7000:~/Documents/lab2$ gedit Add.h
student@student-OptiPlex-7000:~/Documents/lab2$ touch Subtract.h
student@student-OptiPlex-7000:~/Documents/lab2$ gedit Subtract.h
student@student-OptiPlex-7000:~/Documents/lab2$ touch Multiply.h
student@student-OptiPlex-7000:~/Documents/lab2$ gedit multiply.h
student@student-OptiPlex-7000:~/Documents/lab2$ touch main.cpp
student@student-OptiPlex-7000:~/Documents/lab2$ gedit main.cpp
student@student-OptiPlex-7000:~/Documents/lab2$ gedit Makefile
```

```
student@student-OptiPlex-7000:~/Documents/lab2$ make
g++ main.cpp Add.cpp Subtract.cpp Multiply.cpp -o hello
/usr/bin/ld: /tmp/cckJPjU0.o: in function `main':
main.cpp:(.text+0x76): undefined reference to `subtract(int, int)'
collect2: error: ld returned 1 exit status
make: *** [Makefile:2: hello] Error 1
student@student-OptiPlex-7000:~/Documents/lab2$ make
g++ main.cpp Add.cpp Subtract.cpp Multiply.cpp -o hello
student@student-OptiPlex-7000:~/Documents/lab2$ make run
./hello
Enter two numbers: 4
2
Addition result is: 6
Subtraction result is: 2
Multiplication result is: 8
student@student-OptiPlex-7000:~/Documents/lab2$

student@student-OptiPlex-7000:~/Documents/lab2$ make clean
rm -f hello
student@student-OptiPlex-7000:~/Documents/lab2$
```

